

After the Rain

Introduction:

Students will investigate how different ground surfaces around their school correspond to rainfall and explore the influences of stormwater runoff. Over a six-week period, students will monitor three types of surfaces: an impervious surface, a maintained lawn, and a planted garden bed.

At each monitoring station, students will measure soil moisture, surface runoff rate, runoff turbidity, and runoff pH. All groups will contribute their measurements to a shared class dataset so the class can compare how surface type, location, time, and weather conditions affect runoff and water quality

Learning Objectives:

- Explain surface permeability and distinguish between infiltration and surface runoff.
- Explain why pavement produces more runoff than soil or vegetation
- Describe how stormwater runoff collects pollutants from roads, parking lots, and maintained landscapes
- Define turbidity and explain what high and low turbidity indicate
- Develop a research question, hypothesis, and presentation based on findings

Materials (Teams of 4-5 students):

Material	Quantity
Student data sheet or notebook	1
Pencil or pen	1-2
Soil moisture meter	1
Cardstock runoff frame at least 3 feet long	1
Measuring container with 500 mL of water	1
Timer or stopwatch	1
Container for collecting runoff water	1
Turbidity tube with Secchi disk pattern	1
Soil or water test tube for pH testing	1
pH test strips	As needed
pH color chart or test-strip bottle	1
Phone, tablet, or camera for photographs	1

Instructions (Slides):

1. Present slides 1-8 as guided by slide instructions
2. Introduce the research project on slide 9.
3. Present Slide 10 “Setting Up Monitoring Stations”
 - a. Station 1: Impervious Surface
 - i. Select an area of pavement, blacktop, or concrete
 - b. Station 2: Maintained Lawn
 - i. Select a grassy area on school grounds
 - c. Station 3: Planted Garden Bed
 - i. Select a vegetated area with loose soil and established plant roots
4. After each visit, students should enter one row for each station into the class spreadsheet
5. After the final field visit, each group should independently review the completed class data and form a research question
6. Each group will then create a final presentation containing the following sections
 - a. Research Question and Hypothesis
 - b. Findings
 - c. Conclusion
 - i. Whether the hypothesis was supported
 - ii. What the data suggests
 - iii. Which surface handled water most effectively
 - iv. Which surface produced the greatest runoff
 - d. Recommendation
 - i. Students should propose one specific and actionable change that the school or a local government body could take to reduce stormwater runoff

Weekly Field Visit Procedure

1. Record the date, time, group number, and station number
2. Record recent weather or rainfall conditions
3. Visit all three stations
4. Take the soil-moisture reading at the location
5. Conduct the 500 mL runoff test
6. Record the absorption or runoff time
7. Collect runoff water when available
8. Measure turbidity and runoff pH
9. Record observations then enter data into shared class spreadsheet

How to collect weekly measurements:

1. Soil Moisture
 - a. Locate the marked measurement spot at the station.
 - b. Insert the metal probe of the soil-moisture meter fully into the ground.
 - c. Keep the probe in the same marked location used during previous visits.
 - d. Wait ten seconds for the reading to stabilize.
 - e. Record the reading exactly as shown on the meter.

- f. Do not estimate or round unless instructed by the teacher.
- g. Remove the probe carefully and wipe it clean before moving to another station.

For an impervious surface where the probe cannot be inserted, record the reading as **not applicable** unless a specific nearby soil-measurement point has been designated.

2. Surface Runoff Rate

- a. Place the rectangular cardstock frame on the station surface.
- b. Position the frame so that water will remain within the testing path.
- c. Measure exactly 500 mL of water.
- d. Assign one student to pour and one student to operate the timer.
- e. Pour the full 500 mL at one end of the frame.
- f. Start the timer as soon as pouring begins.
- g. Observe the water as it moves across or into the surface.
- h. Stop the timer when the water has either:
 - i. Completely absorbed into the surface, or
 - ii. Completely run off the testing area.
- i. Confirm that no visible pooling remains.
- j. Record the time in seconds.
- k. Record the result as:
 - Absorbed
 - Ran off
 - Mixed result
- l. Add a brief observation if water pooled, escaped the frame, or moved unevenly.

3. Turbidity

- a. Pour the collected runoff water into the turbidity tube.
- b. Hold the tube vertically with both hands.
- c. Point the tube toward a natural light source, such as the sky or a window.
- d. Look straight down through the top of the tube.
- e. Find the Secchi disk pattern printed at the bottom.
- f. Slowly pour water out of the tube.
- g. Stop when the Secchi disk pattern first becomes clearly visible.
- h. Record the remaining water depth in centimeters.
- i. Do not record only “clear” or “cloudy”; record the numerical depth.
- j. Add an observation about the water’s visible color or particles.

4. Water pH

Use the collected runoff-water sample.

1. Pour enough runoff water into the test tube to cover the testing portion of the pH strip.
2. Dip the pH strip into the sample for three seconds.
3. Remove the strip.
4. Hold it horizontally while the color develops.
5. Compare the strip with the chart on the test-strip bottle.

6. Record the closest pH value.
7. Note whether the measurement falls:
 - Below 6.5
 - Between 6.5 and 8.5
 - Above 8.5

Data Sheet (for Students)

Weekly Monitoring Data

Student Name(s):	
Team Number:	
Week:	
Date:	
Time:	

Recent Weather or Rainfall Conditions:

Station	Surface Type	Soil Moisture Reading	Runoff or Absorption Time (seconds)	Result: Absorbed, Runoff, or Mixed	Turbidity Depth (cm)	pH
1	Impervious surface					
2	Maintained lawn					
3	Planted garden bed					